Bike-Sharing Case Study: Should We Just Buy More Bikes?

Sigmoid Analytics case study · Capital Bikeshare, Washington DC, 2011-2012 · all numbers computed by `build_report.py` from `data/`

Executive summary

> **Recommendation: do not buy bikes to fix the complaints. Rebalance ahead of the commute peaks using the demand forecast, and test that with a pilot before any capex.**

- **Timing, not total volume, is what differs.** On working days 58.0% of daily rentals fall in the seven commute hours (07-09, 16-19) against 34.2% on non-working days (Welch p < 1e-100, replicated in both years). Total daily rentals do not differ reliably (p = 0.071).
- **The fleet does not look like the bottleneck at system level, under stated assumptions.** With 20-minute trips, the busiest 1% of hours in 2012 had about 278 bikes on the road, against a reference fleet of about 1,100 (25%). The single busiest hour was about 326. Both the trip length and the fleet figure are external assumptions and are sensitivity-tested (section 4a).
- **Demand can be forecast.** Gradient boosting trained on Jan 2011 to Jun 2012 forecasts Jul-Dec 2012 (never seen) with MAE 42.7 rentals/hr, R^2 0.91 and WAPE 17%. That is 60% lower error than the average-by-hour-and-day-type baseline.
- **Planning for peaks:** a calibrated buffer (x1.36 at rush hours, x1.45 otherwise) covers 90% of all test hours and 95% of rush hours.
- **Known weakness:** the model under-predicts the busiest hours (98% of the top 5% hours, by 124 rentals/hr on average), because demand kept growing and tree models cannot extrapolate.
- **Prototype:** an Operations Assistant (Streamlit) answers plain-English questions with numbers from the trained model and says when the data cannot answer.

What this does not prove: the data records completed rentals only, with no stations, trip durations, empty docks or failed attempts. It supports "timing is the more likely problem", not "stockouts are caused by timing". The pilot in section 4b is what would prove it.

1. Problem framing and assumptions

The CEO's question: **is supply short in total, or short in the wrong place at the wrong time?** The two answers lead to opposite investments, so the analysis is built to tell them apart.

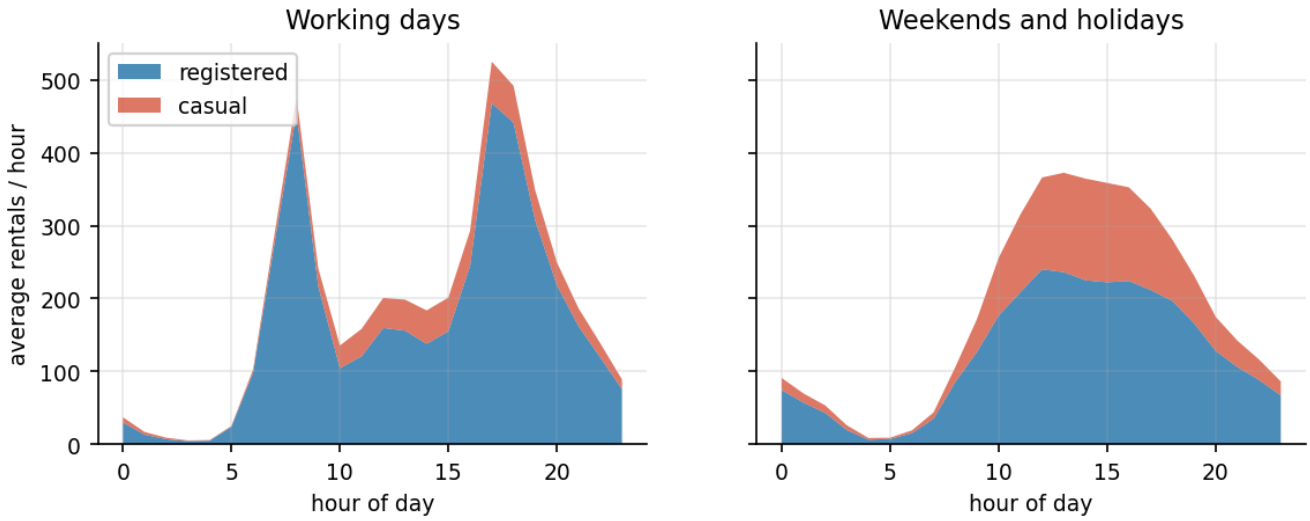
#	Assumption	Why it is reasonable / how to validate
A1	`cnt` counts rentals started per hour, city-wide	Readme; `cnt = casual + registered` on all 17,379 rows
A2	**Observed rentals = served demand** (censored by availability)	A rider who finds no bike is not recorded. Unmet demand is invisible, so dock-status logs are needed (section 4b)
A3	Average trip length **20 minutes**	Not in the data. Tested from 10 to 45 minutes. Replace with trip-level data
A4	Reference fleet **~1,100 bikes**	External figure (Capital Bikeshare, April 2011) quoted in a reference report supplied with the task; **not verified here** and the fleet grew in 2012. Replace with the company's real fleet
A5	Observed weather stands in for forecast weather	In production a weather forecast would be used, so errors would be somewhat higher
A6	One city-wide series stands in for station dynamics	No station ID. Rebalancing routes need station data; the system-level question can still be answered
A7	Planning utilisation ρ is a business choice	Shown for 0.3 to 0.6; better rebalancing allows a higher ρ
A8	Future dates are forecast at the 2012 demand level	The model has never seen a later year; retrain monthly in production

2. Data quality checks

Check	Finding	Action
Nulls / duplicates	0 nulls, 0 duplicate rows in 17,379 rows	none needed
`cnt = casual + registered`	holds on all 17,379 rows	casual and registered are leakage: excluded from features
Hourly vs `day.csv`	hourly sums equal daily totals on all 731 days	used the hourly file; daily file for the day-level test only
Missing hours	165 of 17,544 hours absent, across 76 days	mostly isolated night hours (no rentals, not recorded); storm days (Irene Aug 2011, Sandy Oct 2012, Jan 2011 snow) flagged, not imputed
Humidity = 0	22 rows, all on 2011-03-10	physically impossible: imputed from neighbouring days, not deleted
`atemp` fault	On 2012-08-17 feels-like is constant at 0.24 (about 12 °C) while `temp` is 0.62 to 0.86 (about 25 to 35 °C)	sensor fault; `atemp` is not used by the model (it correlates 0.99 with `temp`), so no impact
Wind speed = 0	12.5% of rows	probable sensor floor; kept, with a flag
`weathersit` = 4	3 rows	merged into category 3
Outliers	kept	the high hours are the real commute peaks that planning must cover
Growth	mean rentals/hr 143.8 (2011) to 234.7 (2012), up 63%	drove the choice of a chronological split (section 5)

3. Feature understanding and engineering

- **Target:** `cnt`. **Excluded:** `casual`, `registered` (sum to the target), `instant` (row index, a hidden time trend), raw `dteday`.
- **Calendar:** hour, weekday, month, season, year, working day, holiday, weekend, plus sine and cosine of hour and month so 23:00 sits next to 00:00.
- **Regime flags:** morning peak (07-09), evening peak (16-19), rush hour, night. Working days have two commute peaks; weekends have one midday hump. They behave like two different demand processes.
- **Weather:** category, temperature, a squared temperature term (demand is not linear in temperature), humidity, wind.
- **No lagged demand** in the final model: a same-hour-last-week feature did not improve the result in testing, and lags would stop the assistant answering what-if questions such as "Friday 6 PM, rainy".
- **Operational conversion:** bikes on the road ≈ rentals per hour × trip minutes ÷ 60 (Little's law). This turns rentals into bikes, the quantity the CEO cares about.



Working days are two-peaked (commute); weekends and holidays have one leisure hump. Mean rentals/hr at 8 AM: 605 on 2012 working days.

EDA findings. Registered riders are 81% of rentals (87% on working days). Casual riders are 32% of weekend rentals and are more weather-sensitive: in light rain or snow casual demand falls to 0.40 of clear-weather level against 0.58 for registered. Mean rentals by weather class: 205 (clear), 175 (mist), 112 (light rain/snow). Humidity correlates -0.32 with demand and is tied to weather. Season means: 111, 208, 236, 199 rentals/hr.

4. Hypothesis and experimental design

> **H1: the complaints are a when-and-where problem, not a how-many problem. Demand is concentrated into two commute windows, and the fleet is only a fraction used even at the busiest hour, so moving bikes before the peaks should reduce shortages more cheaply than buying bikes.**

4a. In-data tests

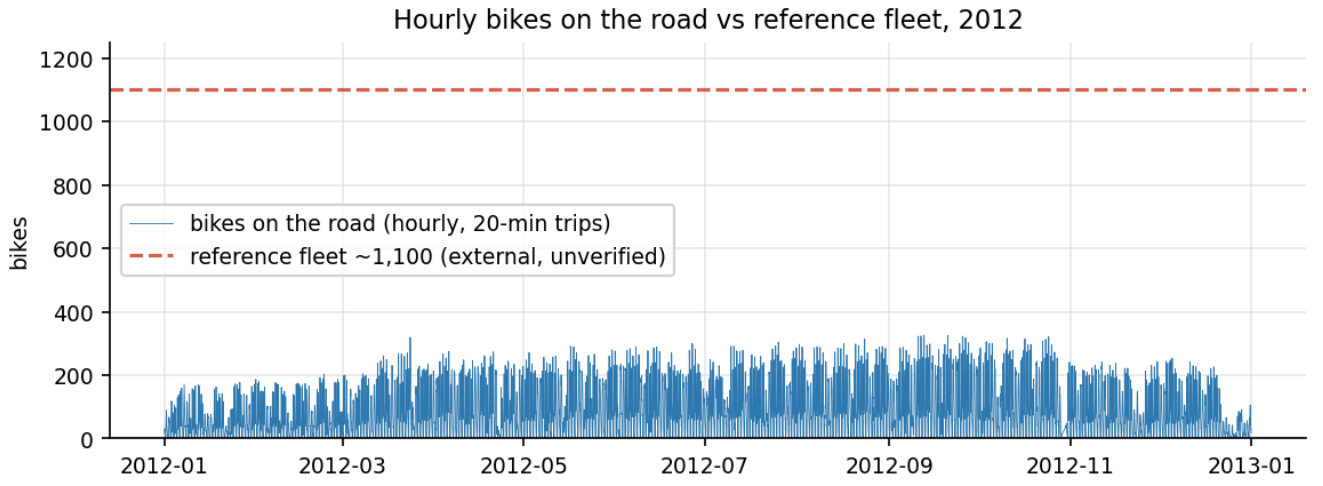
Test 1: timing. H0: the share of daily rentals in commute hours is equal on working and non-working days. Chosen on 2011 and replicated on 2012.

Sample	Working	Non-working	Difference (95% CI)	Welch p	Mann-Whitney p
2011 (n=246 / 113)	0.579	0.341	0.238 (0.228 to 0.247)	< 1e-100	1e-51
2012 (n=248 / 116)	0.581	0.343	0.238 (0.229 to 0.248)	< 1e-100	1e-51
Both years (n=494 / 229)	0.580	0.342	0.238 (0.231 to 0.245)	< 1e-100	< 1e-100
Daily total rentals	4,632	4,344	288 (-23 to 599)	0.071	0.084

The effect is identical in both years (0.238), holds under a rank-based test, and total volume is not reliably different. The standardised effect (d ≈ 5.5) is very large because each day's share has little noise; 100 of 494 working days still have a lower peak share than the highest non-working day, so the groups overlap a little.

Test 2: is the fleet binding? If fleet size were the constraint, peak utilisation would approach 100%. H0: 99th-percentile utilisation ≥ 85%. Day-level block bootstrap (1,000 resamples of whole days).

Year	P99 bikes on the road (20-min trips)	95% CI	Max hour	P99 utilisation of 1,100	Bootstrap p
2011	185	180 to 188	217	17%	< 0.001
2012	278	273 to 283	326	25%	< 0.001



Sensitivity. Utilisation of the single busiest 2012 hour by trip length: 10 min → 15%, 15 min → 22%, 20 min → 30%, 30 min → 44%, 45 min → 67%. It stays well below 100% across the grid, but a long enough trip length would change the picture, which is why trip data matters.

Limitation. This argues against a *system-wide* shortage only. Demand is censored and there is no station dimension, so it cannot show riders turned away at empty docks or prove that rebalancing fixes them. That is what the experiment is for.

4b. Field experiment: randomised pilot (8 weeks)

Element	Design
Unit	Station cluster (about 10 neighbouring stations) × week; clusters contain walk-to-next-dock spill-over
Arms	**A** control (usual rebalancing) · **B** same fleet plus forecast-driven pre-peak rebalancing (fill residential docks before 07:00, downtown docks before 16:00) · **C** (optional) about 15% extra bikes added, to answer the buy question directly
Assignment	Random, stratified by cluster type (residential / downtown / leisure); arms rotate weekly so weather and season balance
Primary metric	Empty-dock minutes per station during commute peaks (needs dock-status logs)
Secondary	Failed rental attempts, completed peak rentals, full-dock minutes, cost per avoided stockout
Guardrails	Midday and off-peak availability, app complaints, operations cost
Analysis	Cluster-robust difference in means or mixed model, adjusting for weather and weekday
Decision rule (fixed in advance)	If B reduces empty-dock minutes about as much as C at lower cost → scale rebalancing and do not buy bikes
Sample size (proxy)	Using 07-09 working-day rentals in 2012 (mean 1,273, SD 324) at 5% significance and 80% power: about 102 working days per arm for a 10% effect, 407 for 5%. Recompute once real empty-dock data exists

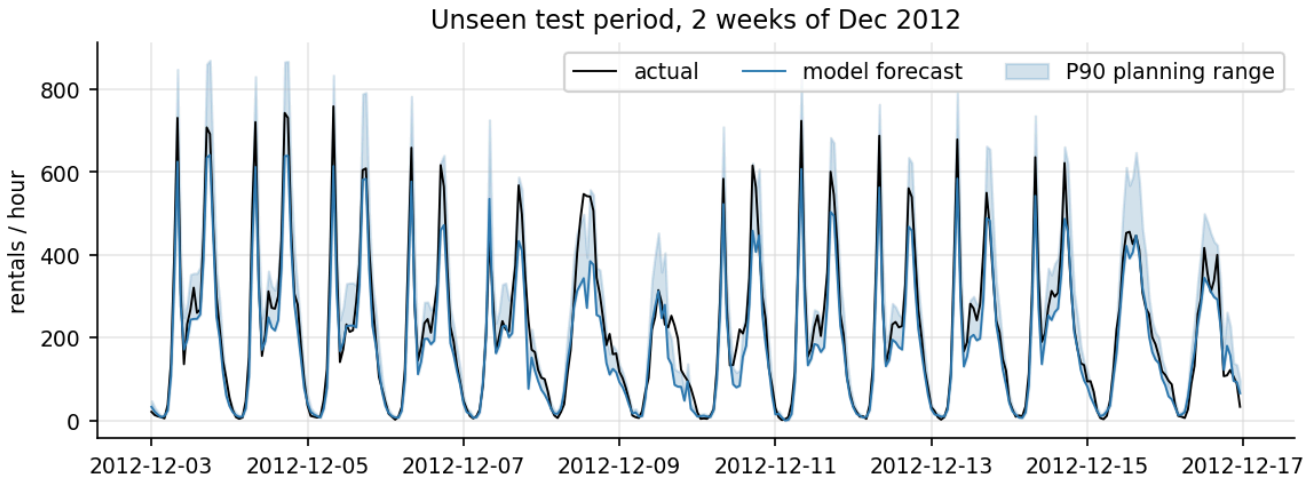
5. Model: how many bikes are needed

Setup. Chronological split, no shuffling: train Jan 2011 to Jun 2012, test Jul to Dec 2012 used once for the final comparison. Model choice used rolling time-ordered validation (Mar-Jun 2012) inside the training period only. Metrics: MAE, RMSE, R^2, WAPE (not MAPE, which explodes on hours with 1-5 rentals), and rush-hour MAE, because under-predicting a peak is the costly error.

Model (test = Jul-Dec 2012)	MAE	RMSE	R^2	WAPE	Rush-hour MAE
Overall mean	177.4	236.1	-0.15	69.8%	362.3
Average by hour and day type	107.2	150.3	0.53	42.2%	218.6
Linear regression	81.6	108.1	0.76	32.1%	140.6
Random forest	50.2	74.7	0.88	19.7%	101.9
Gradient boosting (chosen)	42.7	65.9	0.91	16.8%	81.2

- Gradient boosting wins on every metric; its error is 60% below the average-by-hour-and-day-type baseline an operations manager could compute without a model.
- **Why chronological:** a random 70/30 split gives MAE 22.8, which looks twice as good but lets neighbouring hours of the same day sit in training and test. Training on 2011 and testing on 2012 gives MAE 89.5 and R^2 0.64, because the model never sees the 63% growth.
- **What drives predictions** (shuffling a feature and measuring the rise in test MAE): `hr` +64, `rush_hour` +38, `hr_cos` +28, `hr_sin` +27, `workingday` +11.
- **Planning buffer.** Forecast × 1.36 (rush hours) or × 1.45 (other hours), calibrated on validation folds, covers 90.2% of all test hours and 95.3% of rush hours. A quantile-loss gradient boosting aimed at P90 was tried and covered only about 72% of test hours, so it was rejected.

- **Weaknesses, measured:** 81% of test rush hours are under-predicted (mean bias -50/hr); the busiest 5% of hours by 124. Extreme weather (hurricane days) is not modelled.



From forecast to fleet (indicative)

Fleet needed ≈ P95 of peak-hour bikes on the road ÷ ρ. For working-day rush hours in Jul-Dec 2012, P95 is about **289 bikes on the road** at 20-minute trips.

ρ (max planned peak utilisation)	Bikes needed	If demand grows another 63%
0.3	964	1,573
0.4	723	1,180
0.5	578	944
0.6	482	786

Compared with the unverified ~1,100 reference fleet these are lower, but they move one-for-one with the trip-length assumption. This is not a fleet-size answer: that needs station stock, repairs and trip data.

6. Prototype: Bike Demand Intelligence, Operations Assistant

Goal: let an operations manager ask plain-English questions and get numbers from the trained model, not from a language model. No paid API is used.

`question → intent parser (rules) → trained model + 2011-12 history → answer + chart + how the question was read + assumptions`

Question type	Example
Forecast for a date and hours	How many bikes are expected to be rented tomorrow at 8 AM?
Peak hours	What are the peak-demand hours on working days?
Weekday vs weekend	How does demand change between weekdays and weekends?
Weather effect	What happens to demand during rainy weather?
Highest predicted hours	Which hours have the highest predicted demand?
Evening rush	What is the expected demand during the evening rush hour?
Prediction vs history	How does predicted demand compare with historical demand?
Bikes needed in a scenario	How many bikes are needed at 5 pm on Friday if it rains?
Busiest months	Which months are busiest?

Sample real outputs (run on 2026-11-04, so "tomorrow" is 5 Nov 2026): 8 AM expected **661 rentals** (P90 plan 896), about **220 bikes on the road** (299 at P90) at 20-minute trips; historical November 2012 working-day average at that hour is 602. Light rain or snow cuts the day's forecast by about 49%.

Screenshots of the running Streamlit app

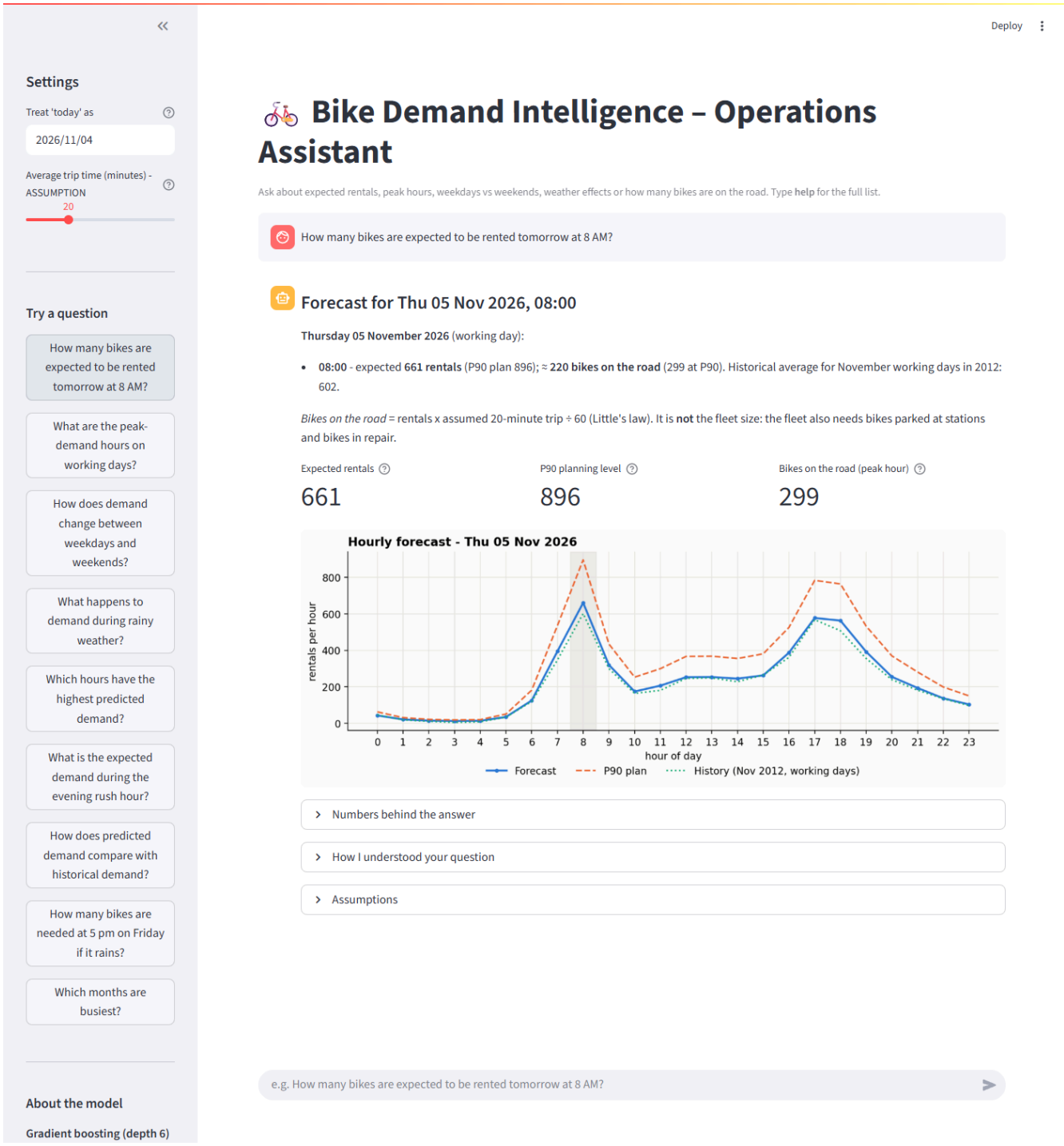


Figure: "How many bikes are expected to be rented tomorrow at 8 AM?" with headline numbers, hourly forecast, P90 plan and 2012 history, and expandable "numbers", "how I understood" and "assumptions" panels.

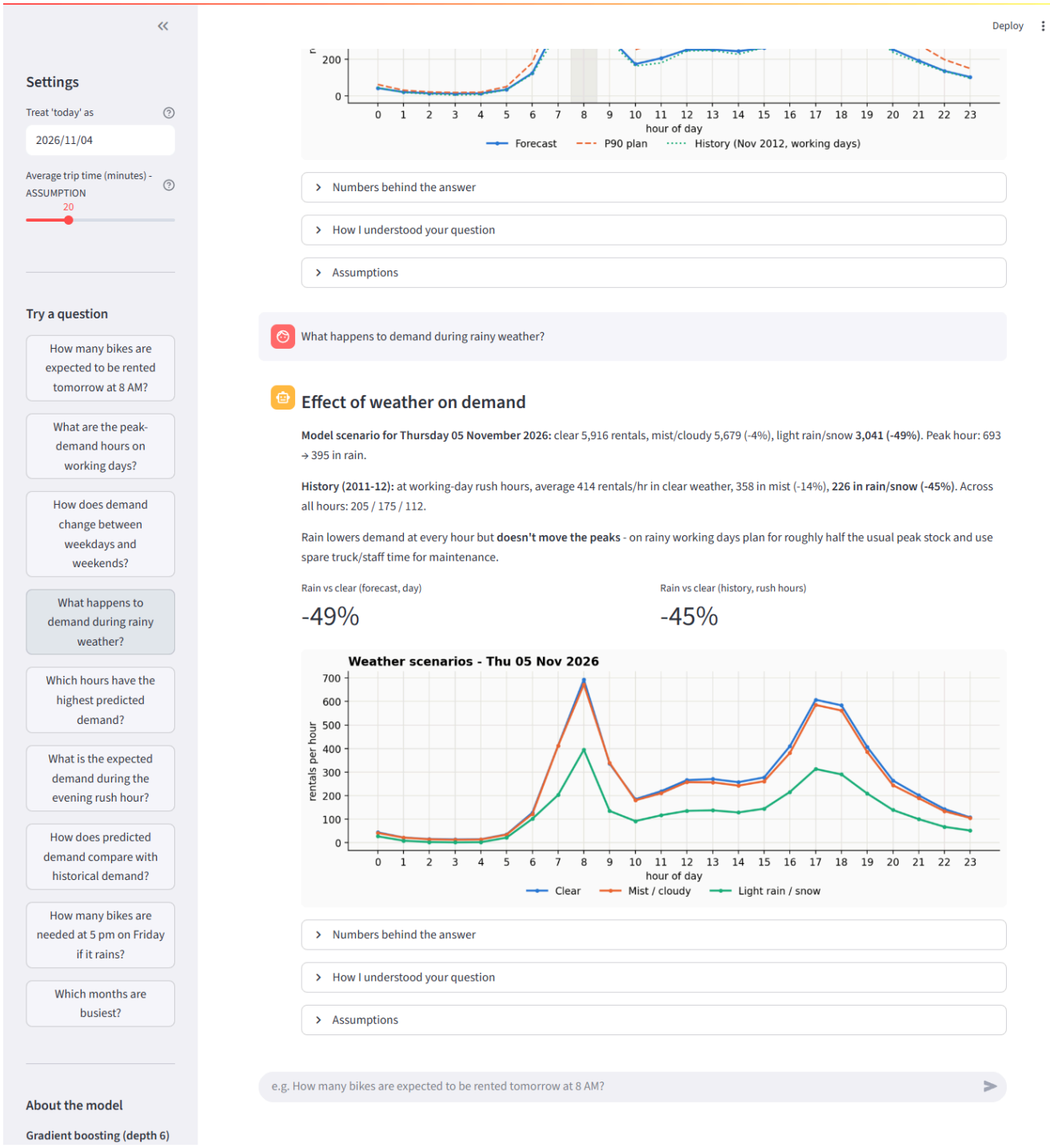


Figure: rainy-weather question.

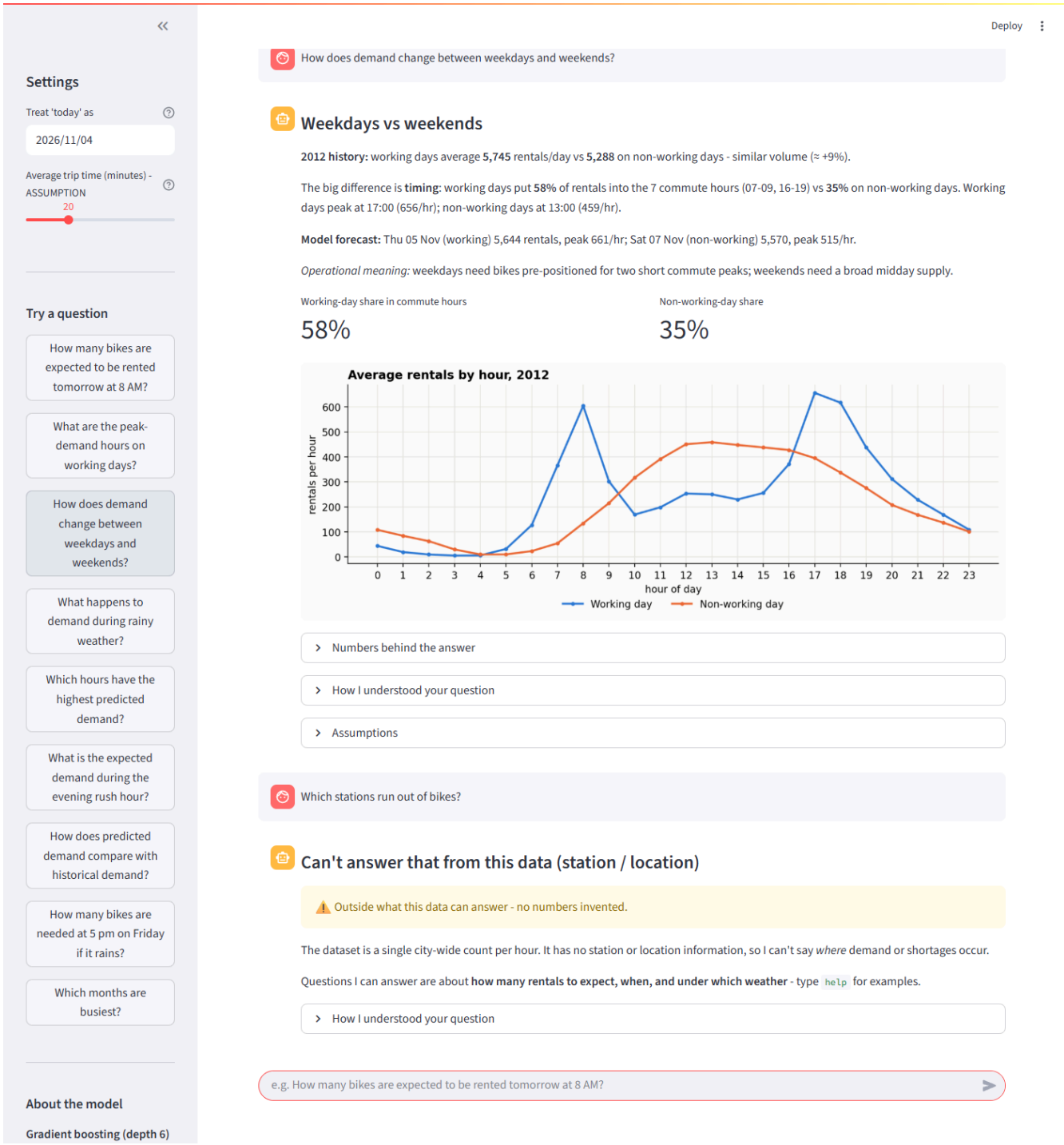


Figure: a question the data cannot answer gets a refusal with the reason, not an invented number.

Design choices. (1) Numbers only come from the model and the data; the parser only routes the question, so answers are auditable. (2) Every answer shows how the question was read and the assumptions used. (3) Questions about stations, revenue, fleet size or stockouts, and anything unrecognised, get a "cannot answer from this data" reply with the reason. (4) Weather not given: it blends that month's usual weather and says so. (5) Future dates are forecast at the 2012 level. Run: `pip install -r requirements.txt` then `streamlit run app.py`.

7. Final recommendation

1. **Do not buy bikes yet.** The evidence points to timing: working-day demand is concentrated into two windows, daily volume is not reliably different by day type, and peak use is a modest share of the reference fleet under stated assumptions.
2. **Forecast-driven pre-peak rebalancing.** Run the forecast each evening and move bikes into commute areas before 07:00 and before 16:00; use the P90 figure for the peak locations. Retrain monthly.
3. **Start logging dock status and failed rental attempts.** This uncensors demand and is the missing evidence.
4. **Run the randomised pilot** (section 4b) with the decision rule fixed in advance.
5. **Buy or lease bikes only if** stations are still empty at the peaks after rebalancing, or for system expansion. Growth of 63% year on year suggests new areas need new bikes, while existing areas need redistribution (this last point is an interpretation, not tested here).

If this is found	Then
Empty-dock minutes stay high after forecast-driven rebalancing	Peak demand exceeds supply in those hubs: buy or lease targeted bikes
Pilot shows no reduction in failed rentals	Reject the timing explanation and reopen the capacity question
Real average trip length is well above 30 minutes	Bikes on the road at peak is much higher; redo the fleet comparison

8. Limitations and next steps

- **Data missing for a fleet decision:** trip start and end times, bikes per station by time, failed rental attempts, station locations, bikes in repair and real fleet size.
- Trip length and fleet size are external or assumed and only sensitivity-tested here.
- Two years of data and a fast-growing system: demand level is non-stationary and tree models hold the last level flat.
- Peaks are under-predicted; the buffer and monthly retraining are the mitigation, not a cure.
- Observed weather replaces forecast weather in testing, so live accuracy would be somewhat lower.
- **Next:** station-level origin-destination flows and dock-status series, availability-adjusted (censored) demand model, then the pilot.

Sources: UCI Bike Sharing Dataset (Fanaee-T and Gama), files supplied with the case; fleet reference figure as quoted in a report supplied by the user, not independently verified.